

# Supplemental Material: Learning Scalar Adjective Intensity from Paraphrases

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## 1 Constructing a Graph of Adjectival Paraphrases

Here we present further detail on the process of constructing JJGRAPH for reproducibility.

### 1.1 Dissecting Paraphrase Pairs

Building a graph to represent paraphrase pairs requires identification of paraphrases with the form  $(RB\ JJ_u \leftrightarrow JJ_v)$  in PPDB-XXL. Paraphrases in PPDB consist of phrase pairs, where each phrase can be one or more words long. Each pair is labeled with a syntactic part of speech; for this work we consider only adjectival phrase (ADJP) paraphrases.

Given a paraphrase pair, we denote as  $P_1$  the phrase with longer token length, and  $P_2$  the shorter phrase. We assume that  $P_2$  consists of a single adjective, and  $P_1$  consists of an adjective modified by an adverb. More specifically, within  $P_1$  of length  $n$ , we identify the adjective as the last token, and the adverbial modifier the concatenated tokens from the first to  $(n - 1)^{th}$  token. For the purposes of this study, phrases where the adverb meets one of the following criteria are ignored: longer than 4 tokens; consists of a single character; consists of the word *not*; ends with one of the tokens *about*, *and*, *in*, *or*, *the*, or *to*; or contains digits.

### 1.2 Bootstrapping Intensifying Adverbs

Recall that building JJGRAPH requires a set of intensifying adverbs in order to select paraphrases of the form  $(RB\ JJ_u \leftrightarrow JJ_v)$  for inclusion in the graph. We used an iterative bootstrapping process to identify intensifying adverbs as outlined in Section 3.1 of the paper.

The process begins with a small seed set of adjective pairs with a known intensity relationship. For the seeds, we identified ad-

jectival phrase paraphrases  $(P_1, P_2)$  in PPDB where the adjective in  $P_1$  is in its base form (e.g., *hard*), and  $P_2$  is either the superlative or comparative form of the same adjective (e.g., *harder*). Such pairs were identified by lemmatizing with NLTK's `WordNetLemmatizer` (Loper and Bird (2002)).

By definition, the base form of an adjective is less semantically intense than both its comparative and superlative forms (e.g., *hard* < *harder* < *hardest*). Thus, the adverb that precedes the base form of the adjective in  $P_1$  is presumed to be an intensifying adverb. We add all adverbs identified in Round 1 of the process to an initial adverb list,  $R_1$ .

Next, we found additional paraphrases where the adverb in  $P_1$  appears in  $R_1$ , but the adjectives in  $P_1$  and  $P_2$  could be anything. For example, where in Round 1 we identified *very* and *pretty* as intensifying adverbs, we found in Round 2 that *pleasant* and *delightful* and that *simple* and *plain* were also related by *very* and *pretty*, respectively. That is, *pleasant* < *delightful* and *simple* < *plain*.

Finally, in Round 3, we identified an additional set of intensifying adverbs by finding paraphrases with the new adjective pairs identified in Round 2. For example, in Round 2 we found the patterns  $[adverb]\ pleasant \leftrightarrow delightful$  and  $[adverb]\ simple \leftrightarrow plain$ , so in Round 3 we found all other paraphrases that fit those patterns (e.g., *more pleasant* < *delightful*, *quite simple* < *plain*). The adverbs in these phrases (e.g., *more*, *quite*) are intensifying. We add all such adverbs identified in Round 3 to an adverb set  $R_3$ , and take as our final set of intensifying adverbs  $R = R_1 \cup R_3$ .

In total, we identified 610 intensifying adverbs using this process. We also carried out a parallel process to identify de-intensifying adverbs, but found that the list of intensifiers was larger, and that the list of de-intensifying adverbs overlapped